Joint Fusion and Conditional Intervention Value Learning for Cooperative Multi-USV Target Defense

Manuscript in preparation

***Abstract*—Cooperative target defense requires the control choices of individual vessels to form a successful team maneuver. We address two distinct decisions in learned–Boids controller fusion: which combination of fusion coefficients the fleet should execute, and when changing a local fusion decision improves the subsequent defense outcome. First, a candidate-interaction model predicts the motion induced by each vessel's admissible actuator mixtures and selects a joint configuration using their pairwise consequences. Second, a conditional intervention value model learns a shared task potential from complete, paired feedback continuations. Single-vessel differences identify useful changes within a joint configuration, while factual outcome calibration and intervention timing guide deployment. In completed development comparisons, the refined joint selector improves success by 10.7 percentage points over independent selection with identical candidates and predictions. The complete method achieves 95.2% success across 856 development episodes, exceeding its frozen fusion predecessor by 9.0 points and a matched ordinary return MLP by 14.4 points. Five independent value fits retain an average 10.2-point advantage over the MLP. These results distinguish joint-selection gains from conditional-value gains; separate held-out, independently retrained baseline, opponent-adaptation, and high-fidelity evaluations determine their final empirical scope.**

***Index Terms*—Multi-USV target defense, cooperative control, controller fusion, joint decision making, conditional intervention value.**

I. Introduction

A fleet defending a target against a faster intruder must coordinate how its vessels move together. Individually plausible maneuvers can send two defenders through the same narrow space, leave an interception direction uncovered, or disturb an otherwise successful approach. The central decision is therefore how available controls should be combined across the fleet, and which changes are worth making as the encounter unfolds. Target guarding and cooperative pursuit provide the task setting [2], [3], [5]; learned marine control provides flexible individual maneuvers [6], [10], [11].

ARBoids [18] blends a learned SAC controller with a Boids controller using an adaptive coefficient. This preserves a structured coordination prior while allowing task experience to change the maneuver. We retain these two control sources and their original physical execution model. Our first contribution concerns the *joint choice of mixtures*: a vessel's preferred coefficient depends on which control candidates its teammates execute. Observing teammate positions alone does not resolve this dependence. We predict the motion associated with concurrent candidate controls and select a combination using the resulting interaction costs.

A joint selection criterion still does not determine whether a local change helps the final task. Avoiding an imminent encounter may delay capture; an early change may help where the same change after the intruder passes the fleet cannot. Our second contribution concerns the *conditional value of changing a fusion decision*. We compare complete feedback continuations that differ in a specified vessel's mixture while retaining the other components of the current joint plan. Both branches encounter the same sampled future disturbances and then follow the same frozen controller on their own evolved states. These comparisons train a common task potential, so a local value difference remains consistent with the value of a complete fleet intervention.

The resulting method connects physical interaction with task consequence. Joint fusion coordinates the current candidate controls. Conditional value learning decides whether to keep that configuration or execute a finite intervention, including a delayed intervention when the opportunity is more favorable. The two mechanisms have separate experimental questions: joint selection is compared with independent selection under the same candidate predictions, and conditional value learning is compared with ordinary absolute-return learning under the same observation, parameter, update, and deployment budgets.

The contributions are:

- A joint fusion rule that resolves interactions between actuator-level learned–Boids candidates while preserving the original control hull.
- A conditional intervention value model that connects paired single-vessel consequences to a shared complete-plan task potential, with a measured factual reference and explicit intervention timing.
- A mechanism-centered evaluation that separates whole-method performance, joint-selection benefit, and the increment over ordinary return learning, with paired failures and successful rescues retained in every comparison.

II. Task and Control Model

A. Dynamics and task outcomes

The numerical environment uses the released ARBoids vessel dynamics and task implementation [18]. A defender obeys

$$\dot{\eta}_i = R(\psi_i)\nu_i, \qquad M\dot{\nu}_i + C(\nu_i)\nu_i + D(\nu_i)\nu_i = \tau_i. \quad (1)$$

The integration period is 0.05 s and the command period is 0.2 s. Each defender thruster remains in $[-500, 1000]$ N;

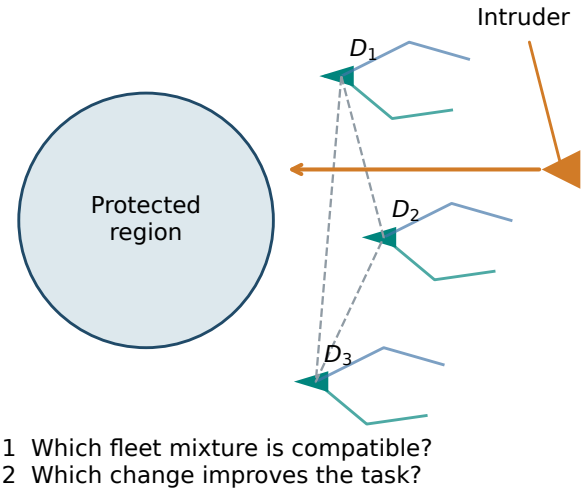


Fig. 1. Target defense and its two decision questions. The fleet must choose mutually compatible mixtures and determine whether a local change improves the eventual outcome. The drawing is schematic; measured trajectories appear in the mechanism and VRX comparisons.

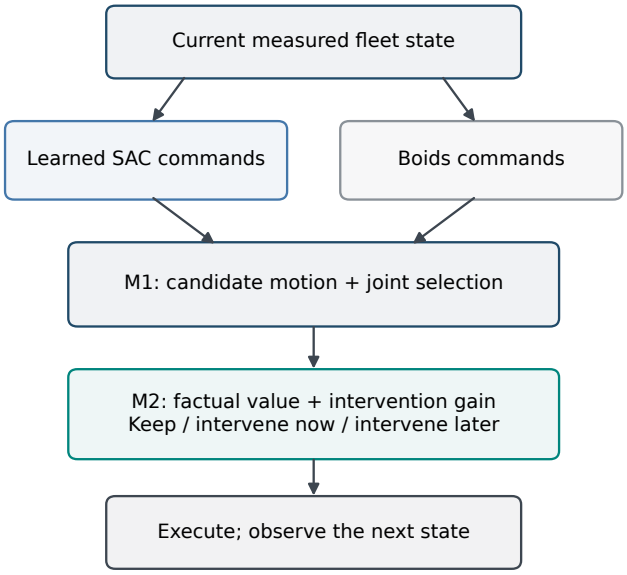


Fig. 2. Execution path. The original learned and Boids controllers produce current commands. Mechanism 1 jointly selects their mixtures. Mechanism 2 evaluates finite interventions against that factual controller; accepted interventions retain the same feedback control sources throughout their execution window.

the agility parameter scales the intruder's thrust bounds. The protected radius is 15 m, and both capture and defender-collision thresholds are 5 m.

We preserve the source termination order. Intrusion failure occurs when the intruder enters the protected radius *or* its distance to the target becomes smaller than that of every defender. Defender collision is checked next, followed by capture, and then successful defense at the 80-s deadline. The radial-pass condition is therefore included in reported source failures, even when the intruder has not physically crossed the target boundary. We additionally record physical boundary crossings, captures, collisions, and episode duration.

The controller observes current vessel geometry, measured velocity, and the available learned and Boids commands. It does not observe future disturbances or the intruder's future control sequence. Source dynamics, source policies, and the source opponent are shared across matched comparisons. The source APF opponent has an order-dependent obstacle update; canonical feature ordering does not change that simulator behavior.

B. Control sources and admissible mixtures

Let u_i^{L} and u_i^{B} be the current learned and Boids thrust vectors. We use two actuator coefficients,

$$u_i = g_i \odot u_i^{\mathrm{L}} + (1 - g_i) \odot u_i^{\mathrm{B}}, \qquad g_i = (g_{i,1}, g_{i,2}) \in [0,1]^2. \tag{2}$$

Allowing separate port and starboard coefficients gives the same admissible representation to joint and independent selectors. Both coefficients remain within the original componentwise control hull. The source adapter supplies a scalar preference, which is mapped to a bounded center $\bar{g}_i$ and shared by the two actuators. Each actuator chooses from $\{0, \bar{g}_i, 1\}$, giving nine candidates per vessel. Duplicate candidates are coalesced.

The three-vessel source SAC policy is the common pretrained control source in the current six-vessel mechanism studies. Its mean teammate embedding permits the observation mapping to other fleet sizes. Training a source policy directly with six vessels is a separate baseline experiment; its results are not replaced by this transfer baseline.

III. Mechanism 1: Joint Selection of Fusion Candidates

A. Candidate motion and pairwise interaction

For every candidate g_i^k, a public-state nominal model predicts the vessel trajectory over 3.2 s at 0.1-s prediction steps. Learned and Boids commands are recomputed every 0.2 s on predicted observations. The candidate coefficient is retained over this forecast, and the executed fleet replans at the physical command rate. This predictor describes the motion associated with candidate intent; it does not replace the complete task continuations used for value learning.

For two candidates, define

$$d_{ij}^{k\ell} = \min_{\tau \in [0,H]} \|\hat{p}_i^k(\tau) - \hat{p}_j^\ell(\tau)\|, \qquad \psi_{ij}(k,\ell) = [7 - d_{ij}^{k\ell}]_+^2. \tag{3}$$

The unary term penalizes departure from the source preference,

$$\phi_i(k) = \frac{1}{2} \sum_{a=1}^{2} \left(\frac{g_{i,a}^k - \bar{g}_i}{0.4} \right)^2. \tag{4}$$

The joint fleet configuration minimizes

$$\boldsymbol{k}^* \approx \arg\min_{\boldsymbol{k}} \left[\sum_i \phi_i(k_i) + 32 \sum_{i<j} \psi_{ij}(k_i, k_j) \right]. \quad (5)$$

The interaction term depends on both vessels' selected candidates. Its role is to distinguish complete combinations that can have similar individual preferences but different physical compatibility.

B. Joint search and the matched independent selector

A width-128 beam constructs complete configurations, followed by three coordinate sweeps. The refined selector additionally refines every retained beam endpoint and performs one pair-block sweep. It accepts a refinement only when both total preference departure and total predicted interaction cost are no worse than the original joint solution. This improves search within the same candidate library and objective.

The independent comparator receives the identical candidates, trajectories, and pairwise cost tensor. With source-preference probabilities $p_j(\ell) \propto \exp[-\phi_j(\ell)]$, it chooses

$$k_i^{\mathrm{ind}} = \arg\min_k \left[\phi_i(k) + 32 \sum_{j \neq i} \sum_{\ell} p_j(\ell) \psi_{ij}(k, \ell) \right]. \quad (6)$$

Thus both methods account for the same predicted candidate interactions. The comparison isolates the consequence of selecting a joint configuration instead of minimizing expected cost independently. Joint optimization has additional search cost, which is reported separately from prediction cost.

IV. Mechanism 2: Conditional Intervention Value

A. A complete intervention and its factual continuation

Let π_0 be the frozen predecessor: either the source fusion controller or the joint fusion controller. An intervention $z = (\boldsymbol{c}, d, h)$ specifies actuator mixtures, delay d, and duration h, measured in physical control ticks. For each vessel, code zero retains the predecessor's decision; the other nine codes specify actuator pairs in $\{0, 0.5, 1\}^2$. Before the delay and after the intervention window, the controller follows π_0. During the window, its learned and Boids commands still update from the current state while the specified mixture is held. Unchanged vessels continue to respond through the predecessor.

For public context x and sampled future ω, let

$$Y(x, z, \omega) = \mathbb{I}\{\text{complete continuation succeeds}\}, \quad (7)$$

$$V(x, z) = \mathbb{E}_\omega[Y(x, z, \omega)]. \quad (8)$$

The factual option z_0 follows π_0 throughout. Two complete options z, z' define a single-vessel comparison when they share timing and all vessel codes except one. Their measured conditional difference is

$$\widehat{\Delta}(x; z, z') = \frac{\sum_r w_r [Y(x, z', \omega_r) - Y(x, z, \omega_r)]}{\sum_r w_r}. \quad (9)$$

Common futures pair the outcomes; later controls are recomputed on each branch's evolved state. Full terminal outcomes include failed as well as successful branches. Sampling may emphasize development failures, but deployed decisions receive only the current context.

B. One task potential for local differences and complete plans

A candidate-conditioned graph network represents a shared scalar potential $V_\theta(x, z) \in [0, 1]$. Node features contain vessel state, both source controls, the factual control, the intervention code, and its physical command change. Pair messages encode relative geometry and nominal response features. Fleet pooling and timing features produce the complete-plan value. Local differences use this same potential,

$$\Delta_\theta(x; z, z') = V_\theta(x, z') - V_\theta(x, z). \quad (10)$$

Consequently, reversing a comparison reverses its sign, and differences sum to zero around a closed comparison cycle. The method does not add independent local scores to approximate a fleet outcome.

The learning objective combines complete outcomes with their conditional differences:

$$\mathcal{L} = \mathcal{L}_Y + 4\mathcal{L}_{\mathrm{local}} + 2\mathcal{L}_{\mathrm{factual}\Delta} + \mathcal{L}_{\mathrm{rank}} + 4\mathcal{L}_{\mathrm{factual}} + \mathcal{L}_{\mathrm{bank}}. \quad (11)$$

Here $\mathcal{L}_Y$ is binary outcome cross entropy, $\mathcal{L}_{\mathrm{local}}$ fits same-future single-vessel differences, and $\mathcal{L}_{\mathrm{factual}\Delta}$ fits differences from the factual option. The ranking term uses outcome-discordant pairs while retaining neutral pairs in its normalization. The last two terms anchor measured factual outcomes and additional states from complete factual trajectories. Sampling corrections retain the chosen comparison population.

Three width-64 models receive 8,000 updates each, for 24,000 total updates and 145,731 trainable parameters. The matched ordinary-return comparator is one three-hidden-layer ReLU MLP with width 234 and 145,783 parameters. It receives the same 66,451 unique terminal observations, derived labels, factual-state observations, candidate catalog, and 24,000 updates. Its objective is pooled absolute-outcome cross entropy and its decision is the maximum predicted return. The graph representation, paired objectives, factual calibration, and local support model are constituents of the proposed mechanism; matching network parameter count does not equate their extra calibration storage or inference cost.

C. Factual calibration, timing, and execution

A measured factual bank estimates the success probability of retaining π_0 in a nearby context. Calibration transports that factual value to the complete neural potential while preserving its learned differences except where the success range $[0, 1]$ requires clipping. Locally measured paired residuals then correct the value differences. Rescue and harm observations, ensemble disagreement, and distance from observed support contribute to the decision penalty.

An intervention that starts after factual termination cannot change that outcome. For delay d, factual trajectory data provide estimates of survival until d, separately for factual success and failure. These estimates bound possible rescue and harm and prevent a delayed option from claiming an opportunity that has already ended. The deployed rule is

$$z^* = \arg\max_{z \neq z_0} \{\widehat{\Delta}(x; z_0, z) - \beta U(x, z)\}, \quad (12)$$

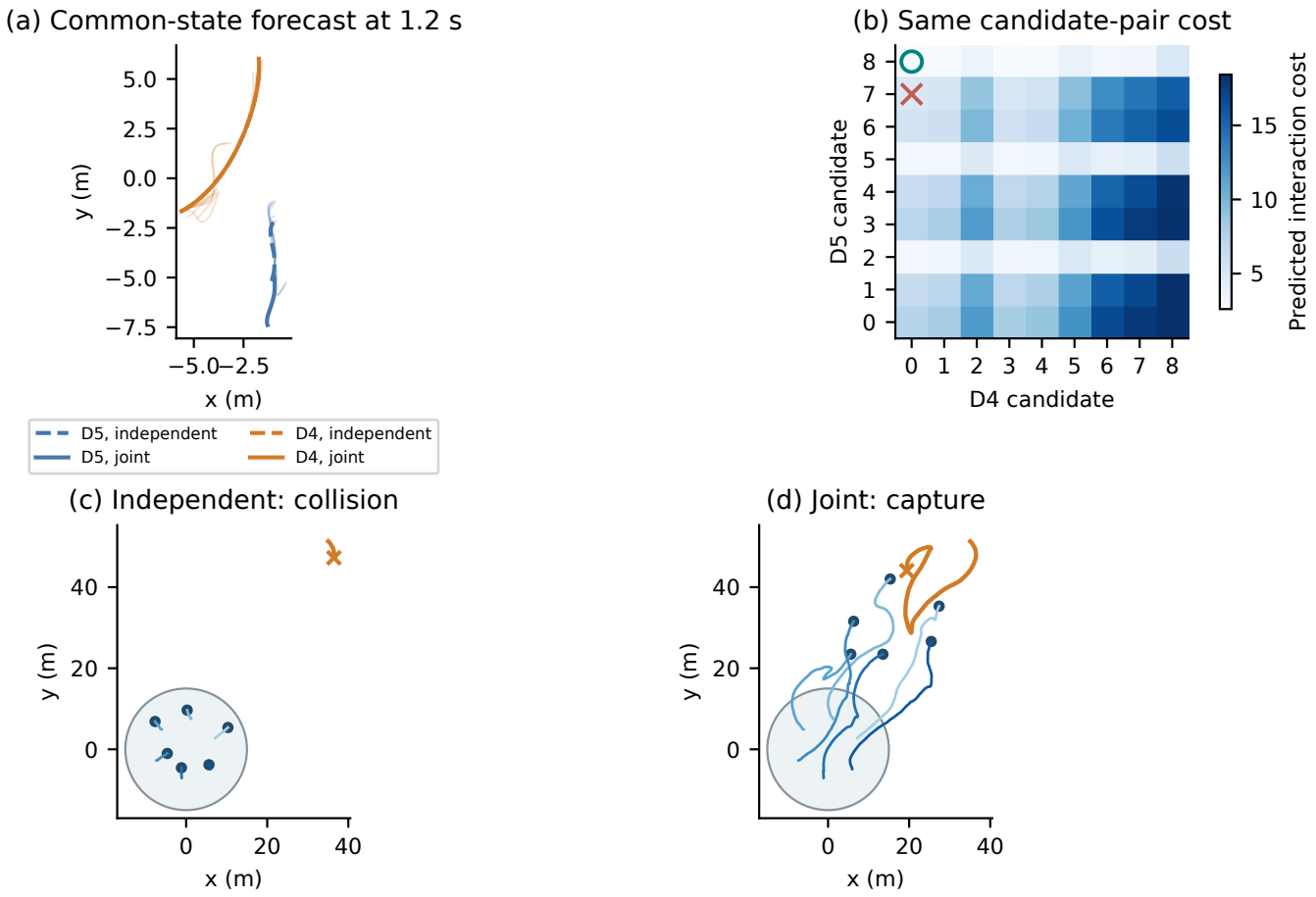


Fig. 3. Mechanism 1 on development scene 431019006. The top panels compare identical candidate forecasts at a common state at 1.2 s on the independent trajectory. The bottom panels replay both complete policies from the same initial state and future: independent selection collides at 1.8 s, whereas joint selection captures at 17.8 s. This illustration is the first joint rescue of an independent collision in the fixed 4310 row order.

and executes z^* only if this score exceeds a positive margin. Otherwise it retains the factual controller and revisits the decision on the next eligible observation. The reference implementation queries every five physical ticks and accepts at most one intervention per episode. The ordinary MLP shares this observation clock and intervention budget.

The uncertainty penalty is a decision quantity, not a guaranteed coverage interval for an unseen state. Development importance weights specify objective mass rather than independent replications. New refits use effective future counts for the rescue/harm penalty. Replacing π_0 requires freshly generated states and terminal labels; outcomes measured under one predecessor are not relabeled as outcomes of another.

V. Experimental Design

The evidence follows three comparisons: the complete method against adequately trained relevant baselines; joint versus independent fusion with matched candidates and predictions; and conditional intervention value versus matched ordinary return learning. Four configurations separate the two mechanisms: F00 uses neither mechanism, F10 uses joint fusion, F01 uses conditional value on the source predecessor, and F11 uses conditional value on the joint predecessor. Each value model is trained with continuations from its own frozen predecessor.

All primary contrasts pair the same initial layout, agility, fleet size, and future seed. Repeated futures from one layout remain in the same bootstrap cluster. Success differences, rescue/harm counts, collision, and capped duration are reported together. Capture-only duration is secondary because successful episodes can differ across methods. Independent learning repetitions are distinguished from additional evaluation seeds and additional future samples.

The current 856-episode development collection contains 256 episodes from the original source suite and 200 episodes in each of three later suites. The source suite repeats 64 layouts under four futures; the other suites contain new layouts. The 4310 suite did not enter the reference value-training data, but its repeated use in model selection makes it development evidence. The reserved 4322 layout family remains outside development and is evaluated only after the method and comparators are frozen.

Five value-learning repetitions use seeds 42–46 with the same frozen source and fusion predecessor. They test value-learning repeatability. Independent source-policy retraining and complete pipeline refitting are separate experiments. Base-

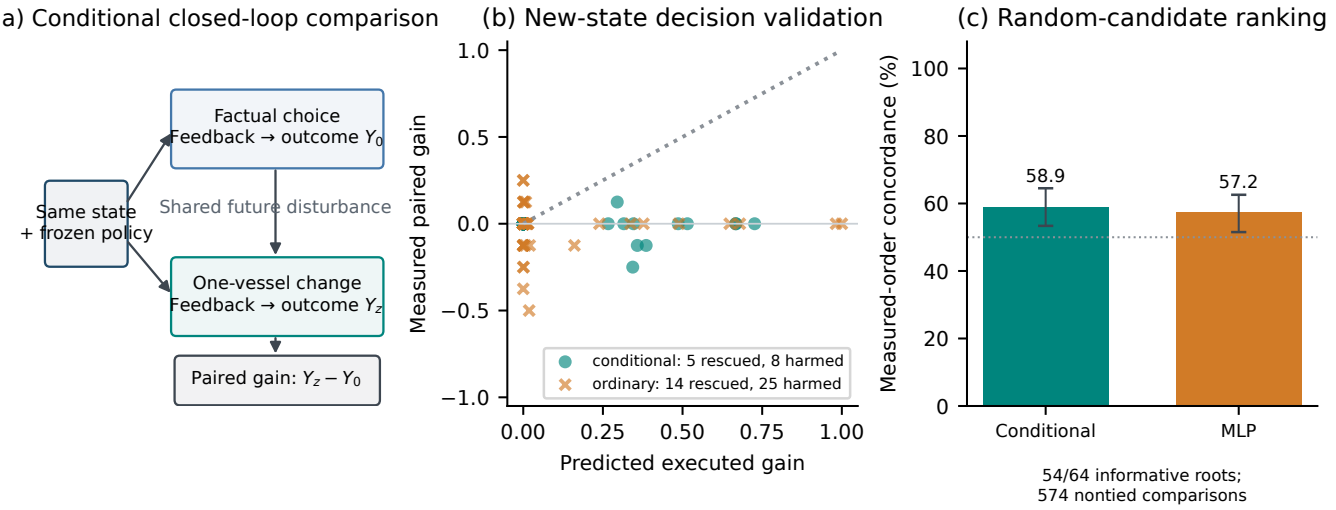


Fig. 4. Mechanism 2. Paired continuations share the initial state, other current decisions, frozen continuation policy, and sampled future. Their eventual outcomes supervise a shared task potential. Factual success, intervention delay, and measured support determine whether the selected change is executed. The ranking panel compares six randomly sampled candidates at each of 64 new states using eight shared futures; error bars are 95% intervals from a stratified bootstrap over states.

TABLE I
COMPLETED DEVELOPMENT RESULTS. COUNTS USE THE SAME PAIRED EPISODES WITHIN EACH COLUMN. THE SOURCE CONTROLLER IS TRANSFERRED FROM THREE-VESSEL TRAINING; THIS IS NOT THE FINAL SIX-VESSEL RETRAINED BASELINE COMPARISON.

Method	Source /256	4310 /200	Earlier /200	Later /200	Total /856
F00: source	37	21	14	25	97
F10: joint fusion	225	171	169	173	738
F01: conditional value	111	57	64	64	296
F11: complete	254	175	196	190	815
Matched ordinary MLP	221	163	157	151	692
Independent fusion	205	153	153	160	671
Refined joint fusion	236	176	172	179	763

line SAC, residual policy, and ARBoids runs retain the source million-step training budget, with the fleet size and actual random seed recorded explicitly.

VI. RESULTS

A. Complete method and the four configurations

The complete reference method succeeds in 815 of 856 episodes (95.2%), compared with 738 (86.2%) for its frozen joint-fusion predecessor. It rescues 87 predecessor failures and harms 10 predecessor successes, adding 9.0 percentage points with a layout-cluster bootstrap interval of [6.9,11.2] points. These development intervals describe resampling uncertainty and do not correct for adaptive model selection. F01 also improves on F00, showing that the value mechanism can recover useful interventions without joint fusion, while the much stronger F11 performance demonstrates the benefit of applying it to coordinated candidate choices.

The refined standalone joint selector obtains 763 successes (89.1%). It is a stronger mechanism-one configuration than the predecessor used to collect the reference F11 data. Its corresponding value model is refitted from fresh complete continuations before making a matched claim about their combination. The two joint configurations are identified separately in the table.

B. Direct evidence for joint selection

The refined joint selector obtains 92 more successes than the independent selector across the same 856 episodes, an absolute gain of 10.7 points (development layout-cluster interval [7.5,14.1]). It rescues 144 independent failures while harming 52 independent successes. Its per-suite increments are 12.1, 11.5, 9.5, and 9.5 points. Because both methods receive the same nine candidates per vessel and the same prediction tensor, this contrast directly measures the value of resolving their combination jointly.

The original source backbone and four independently trained source backbones give five separate joint-versus-independent comparisons on the common 228-episode screen. Joint selection obtains 948/1,140 successes, compared with 843/1,140 for independent selection. The mean gain is 9.2 points; individual gains are 12.3, 4.4, 7.9, 10.1, and 11.4 points. All four additional backbones complete the original one-million-step source training. This experiment repeats the analytic joint selector across independently learned source commands; it does not yet constitute five retrainings of the complete value-learning pipeline.

The strengthened state-attention and candidate-attention alternatives obtain 446/856 (52.1%) and 551/856 (64.4%) successes. Both use two attention layers of width 128, 192,769 trainable parameters, 24,000 optimizer updates, and the same 24,576 measured teacher queries collected on teacher and student trajectories. Training balances changed and unchanged teacher choices and retains every physically equivalent candidate label; candidate trajectories are expressed in a common task frame. The joint selector supplies supervision only during training. At deployment, each replacement selects candidates

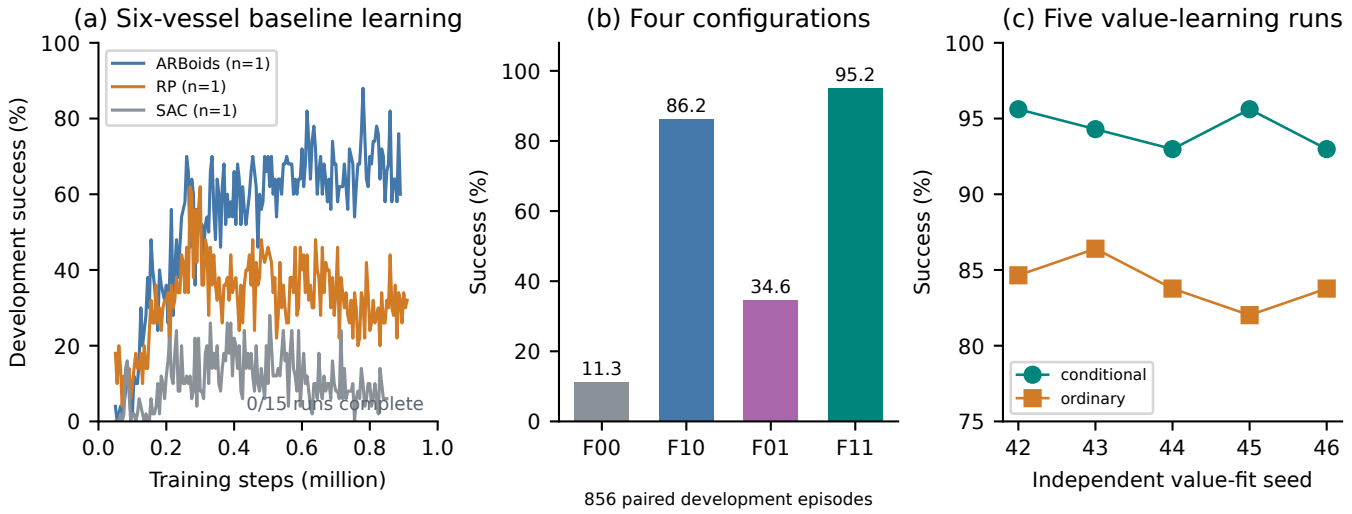


Fig. 5. Overall evidence. Baseline learning curves are shown at their actually completed training steps. Four-configuration success and five independent value fits use paired development episodes. Independent value fits share a frozen source backbone and must not be interpreted as five complete pipeline retrainings.

through its own neural outputs. These fitted alternatives improve substantially over their initial versions but remain below independent selection (78.4%) and joint selection (89.1%). The matched joint-versus-independent contrast provides the direct evidence for joint decision making; the replacement comparison additionally tests whether these neural representations recover its benefit.

C. Conditional value versus ordinary return learning

The complete reference controller exceeds the matched ordinary MLP by 123 successes, or 14.4 points (development layout-cluster interval [11.6,17.4]). The paired outcomes contain 139 rescues and 16 harms. The per-suite gains are 12.9, 6.0, 19.5, and 19.5 points. The smaller 4310 increment is retained because it is informative about generalization beyond the fitting layouts.

Across five independent value fits, the conditional model succeeds in 1,075 of 1,140 paired evaluation episodes and the ordinary MLP in 959, a mean increment of 10.2 points. Each run contains the same 128-source/100-4310 screen, so these are repeated learning outcomes on a shared development collection. The per-run increments are 11.0, 7.9, 9.2, 13.6, and 9.2 points. Seed 44 reverses the advantage on the 4310 subset (85 versus 89 successes); its combined advantage does not erase that adverse result.

Adding paired short-response observations and 6,000 updates to both estimators gives 813/856 successes for conditional value and 684/856 for its matched MLP, a 15.1-point difference with 143 paired rescues and 14 harms. Both refits receive the same 82,857 terminal observations and 30,000 cumulative updates. This refit strengthens the matched-MLP contrast but remains two successes below the reference complete method; its separate VRX gains are reported below.

New-state value validation samples one eligible pre-action state uniformly from each of 64 complete factual trajectories on new layouts. At every root, the conditional model and ordinary MLP propose their own action; both choices and the factual action are evaluated to termination under eight new shared futures. The conditional model intervenes at 12 roots and yields 408 successful continuations, versus 400 for the MLP and 411 for the factual controller out of 512 continuations. Relative to the factual controller, conditional decisions rescue five futures and harm eight; the MLP rescues 14 and harms 25. Conditional value therefore reduces harmful changes, but neither learned selector improves on factual continuation in this random-state experiment. Positive predicted gains frequently correspond to zero measured gain, identifying a remaining calibration and state-coverage weakness. Component controls separately remove delayed choices, temporal survival bounds, and factual calibration while retaining learned parameters and the common deployment budget.

A separate collection measures six outcome-independent random candidates at each of 64 new states under eight shared futures. Of these states, 54 have at least one pair with different empirical success probabilities, yielding 574 nontied comparisons. Conditional value agrees with the measured ordering in 58.9% of comparisons, versus 57.2% for the ordinary MLP; the paired difference is 1.7 points with a stratified state-bootstrap interval of $[-5.2,9.0]$ points. Its gain-prediction root mean squared error is 0.247, compared with 0.402 for the MLP. Thus the conditional estimator improves gain calibration in this collection, while a ranking advantage over ordinary return learning remains unresolved. All candidates and neutral outcomes remain in the calibration calculation; only outcome ties are excluded from the ordering score.

On the 228-episode component screen, the complete model achieves 218 successes, immediate-only choices 214, removal of temporal bounds 218, and removal of factual calibration 215. Relative to immediate-only choices, the complete model rescues eight episodes and harms four. Factual calibration rescues three and harms none. Removing temporal bounds changes neither success nor terminal duration in this screen;

these completed outcomes therefore provide no separate performance gain for that bound. All three controls achieve the same 91/100 successes on the 4310 subset, locating the observed component gains in the source-development subset.

D. Agility, fleet size, and opponent adaptation

Agility comparisons retain the same methods and paired scene design. Cross-fleet inference uses all current vessels in the shared neural potential, projects valid intervention roles to the current fleet, and generates every single-vessel alternative for that fleet size. Six-vessel empirical calibration is not transferred as if it were a measurement at another fleet size.

In the completed 16-scene screens with three, four, five, seven, and eight defenders, the complete method succeeds in 16, 15, 15, 12, and 10 episodes, respectively. The joint predecessor obtains 14, 15, 15, 14, and 10 successes. Conditional value therefore adds two successes at three defenders, preserves the counts at four, five, and eight, and loses two at seven. Refined joint selection obtains 14, 15, 15, 13, and 12 successes, compared with 14, 16, 16, 11, and 9 for independent selection. These paired screens identify where each mechanism transfers and where its current mapping loses performance.

Opponent adaptation uses one common learned attacker trained against a fixed source defender. All defender methods face the same frozen attacker before adaptation. New paired continuations are then collected under that opponent and mixed with identified APF training observations. The same adapted defenders are reevaluated against both the learned attacker and the original APF opponent, so recovery and retention remain separate outcomes.

On the 64 adaptation-development layouts, the complete method improves from 49 to 59 successes. This improvement does not transfer to 64 new paired layouts: the complete method obtains 54 successes before adaptation and 53 afterward, while the matched ordinary MLP obtains 52 in both conditions. Conditional value alone changes from 22 to 20 successes. On the original 856 APF episodes, the adapted complete method retains 805 successes, compared with 815 before adaptation; the ordinary MLP changes from 692 to 673, and conditional value alone from 296 to 281. Thus this adaptation recipe improves its fitting-layout performance but does not establish recovery on new layouts or full retention against the original opponent.

E. High-fidelity closed-loop comparison

The VRX comparison [25] executes the current learned–Boids commands in two settings with agility 2.5 and 3.5 and two scene seeds per condition. Each method runs all eight scenes. VRX uses a 0.2-s control period, 15-m target radius, 5-m collision radius, 5.5-m capture radius, and 100-s time limit. These settings are reported separately from the numerical 80-s task.

All 80 completed trials passed the execution-validity checks. The complete reference method captures in 7/8 scenes and breaches in one; its matched single-intervention MLP captures in 7/8 and collides in one. Refined joint fusion captures in 7/8 and collides in one, the independent selector in 4/8, conditional value alone in 6/8, and the transferred source policy in 2/8. The factual joint predecessor captures in all eight scenes. An earlier ordinary-return comparator with a different predecessor and execution budget also captures in all eight; that result remains separate from the matched comparisons.

After additional paired short-response training, the refitted complete method captures in 8/8 scenes and its matched refitted MLP in 6/8. The two additional captures rescue one collision and one target breach without harming a matched MLP success. The refit also captures the scene where the reference complete method breached, as shown in Fig. 8. These eight development scenes establish a matched-comparator advantage for the refit, while its success count remains equal to the factual joint predecessor. Its mean capture time is 12.9 s versus 9.8 s for the predecessor, so this batch does not establish an additional efficiency gain from mechanism two. The valid reference failure remains in the comparison.

VII. Discussion

The two mechanisms solve related but distinct problems. Joint fusion coordinates physically compatible candidate controls; conditional value assesses whether a finite change improves the eventual task. A large gain over a transferred source policy alone cannot establish both contributions. The matched joint-selection and ordinary-return controls provide the more specific evidence.

The shared potential makes local differences consistent with complete-plan values, but consistency does not guarantee accurate generalization. Factual calibration can prevent avoidable damage near observed states, while timing distinguishes an executable rescue from a change that starts too late. New-state paired continuations and the internal component controls test these roles directly. Inference cost includes the neural potential, factual bank, and paired residual calibration; it is reported in addition to trainable parameter count.

Current development gains are substantial, and their limitations are concrete: gains vary by layout family, one value-fit seed loses to the MLP on the untrained 4310 subset, and the improved VRX value model matches its factual predecessor's success count while taking longer to capture. These findings guide the stronger-predecessor refit and new paired training rather than changing the task definition or omitting adverse outcomes. Final generalization claims use the frozen-method held-out comparison and complete independent training runs.

VIII. Conclusion

Joint fusion and conditional intervention value learning provide two complementary ways to improve cooperative target defense. The first selects compatible fleet mixtures from common learned and Boids candidates. The second learns whether and when a local change improves complete closed-loop outcomes, using paired differences tied to one task potential. Completed development comparisons show a 10.7-point joint-selection gain, a 14.4-point advantage over matched ordinary return learning, and a 9.0-point increment over the complete method's factual predecessor. The remaining empirical scope is determined by independent full-pipeline

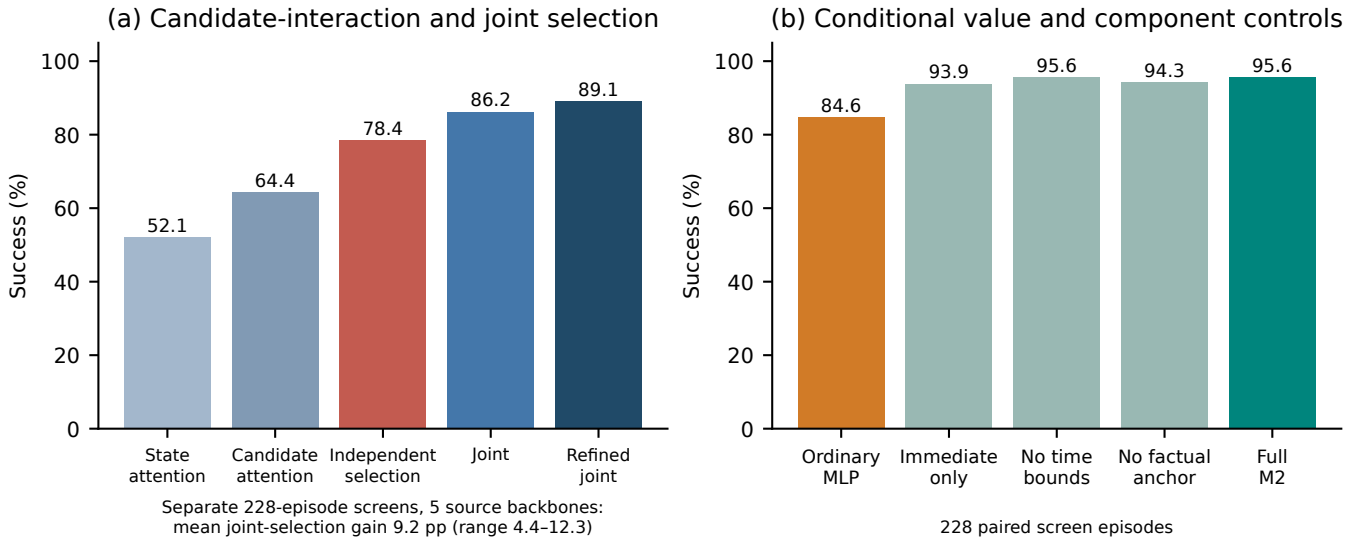


Fig. 6. Direct mechanism comparisons. Left: joint and independent selection share candidate motions and pairwise costs; the two neural alternatives share their measured supervision and learning budget. Right: conditional value and the ordinary MLP share observations, network-parameter budget, update count, factual predecessor, candidate catalog, and intervention budget. Internal controls isolate temporal opportunity and factual calibration.

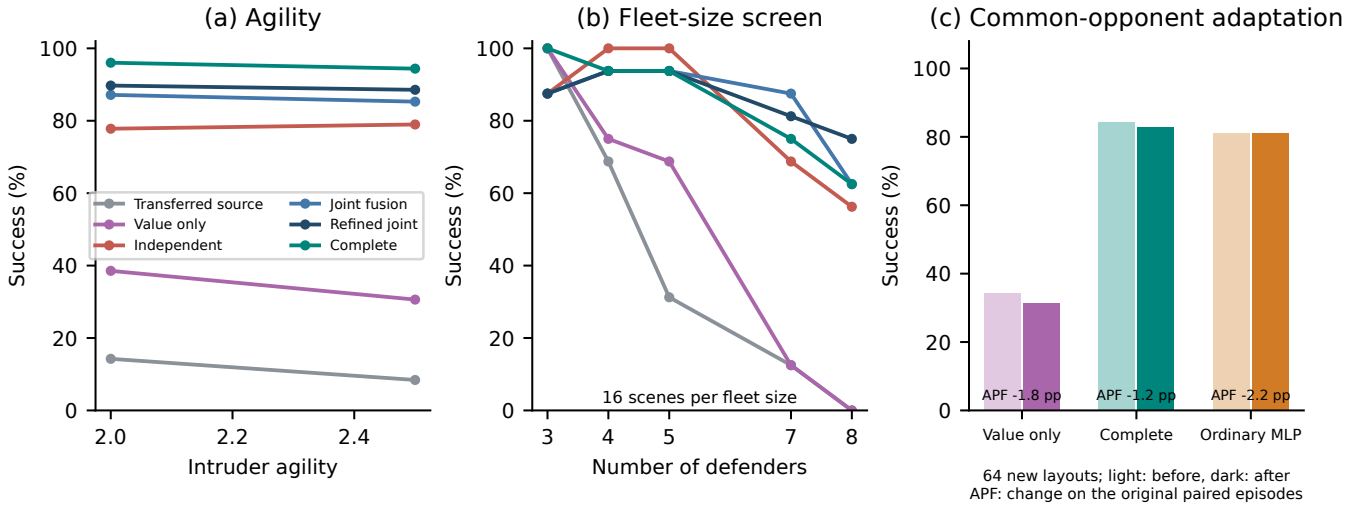


Fig. 7. Scope of the two mechanisms: paired agility comparisons, explicit cross-fleet mappings on 16 scenes per fleet size, and common-opponent adaptation on 64 new paired layouts. The adaptation panel also reports the change in success rate on the original APF episodes.

training, frozen held-out evaluation, opponent adaptation, and the retained high-fidelity successes and failures.

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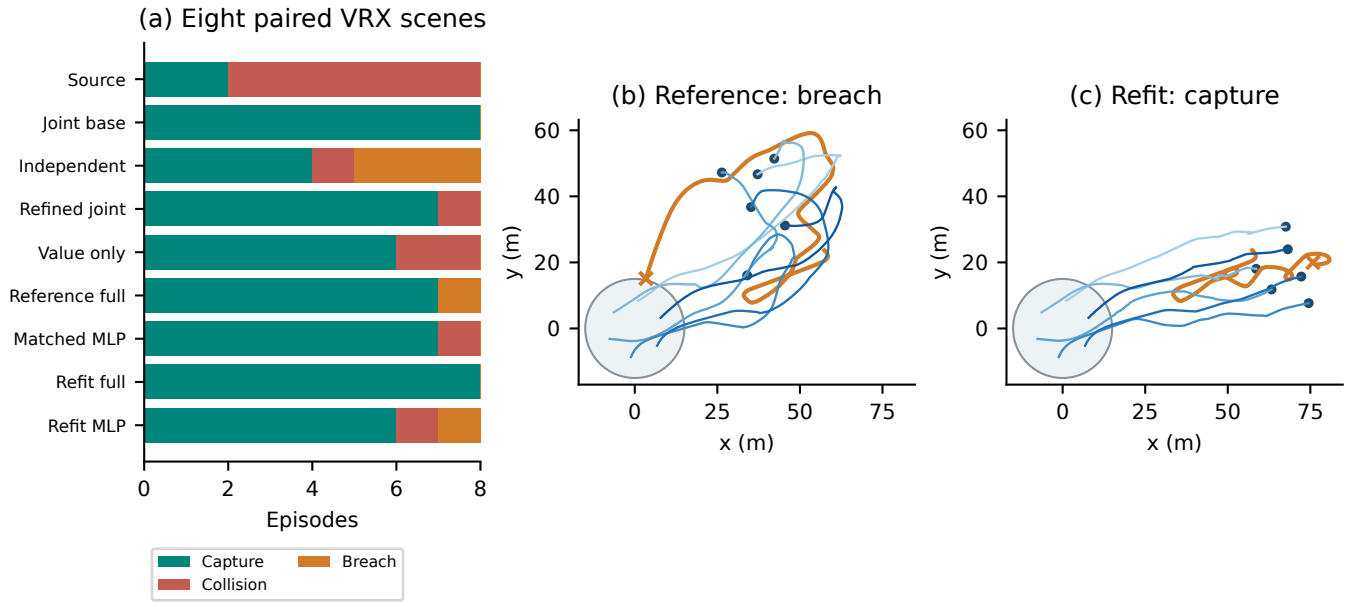


Fig. 8. Eight-scene paired VRX development comparison. Reference and short-response refitted complete methods each have a matched ordinary-return comparator. The trajectories retain the reference method's valid breach and show the refitted method's capture on the same scene.

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